UNIT – II: TYPES OF LASERS AND OUTPUT MODULATION METHODS

Solid State Lasers – Ruby and Nd-YAG Laser – Gas Lasers – He-Ne and CO2 lasers – semiconductor lasers – Heterojunction Lasers – Liquid Dye Lasers – Q switching and mode locking.

Several ways to classify the different types of lasers

- What material or element is used as active medium
- Mode of operation : CW or Pulsed

Classification may be done on basis of other parameters

- Gain of the laser medium
- Power delivered by laser
- Efficiency or
- Applications

Preference to classify the lasers on the basis of material used as Active Medium.

Broadly divided into four categories

- Solid lasers
- Gas lasers
- Liquid lasers
- Semiconductor lasers

TYPES OF LASERS

Based on the type of active medium, Laser systems are broadly classified into the following categories.

S.No	TYPES OF LASER	EXAMPLES
1.	Solid State laser Ruby Laser	Nd:YAG laser
2.	Gas laser He-Ne Laser,	CO_2 Laser, Argon – ion laser
3.	Liquid Laser	$SeOCL_2$ Laser, Europium Chelate Laser
4.	Dye laser	Rhodamine 6G laser, Coumarin dye laser
5	Semiconductor Laser	GaAs laser, GaAsP laser

olid state Laser - NdYAG LASER

It is a solid state and 4 level system as it consists of 4 energy levels. Nd ion is rare earth metal and it is doped with solid state host.Due to doping, yttrium ions get replaced by the Nd^{3+} ions. Also, the doping concentration is around 0.725% by weight.

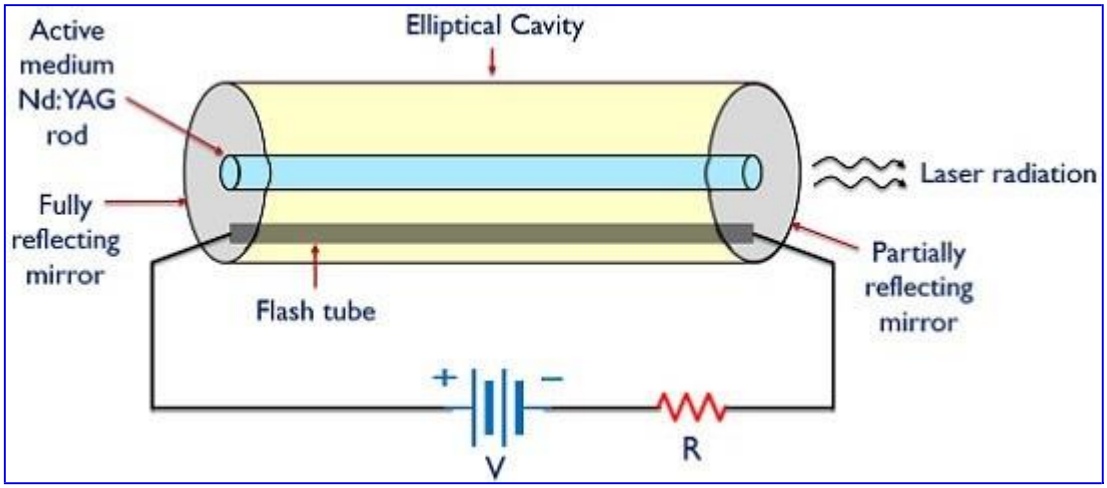
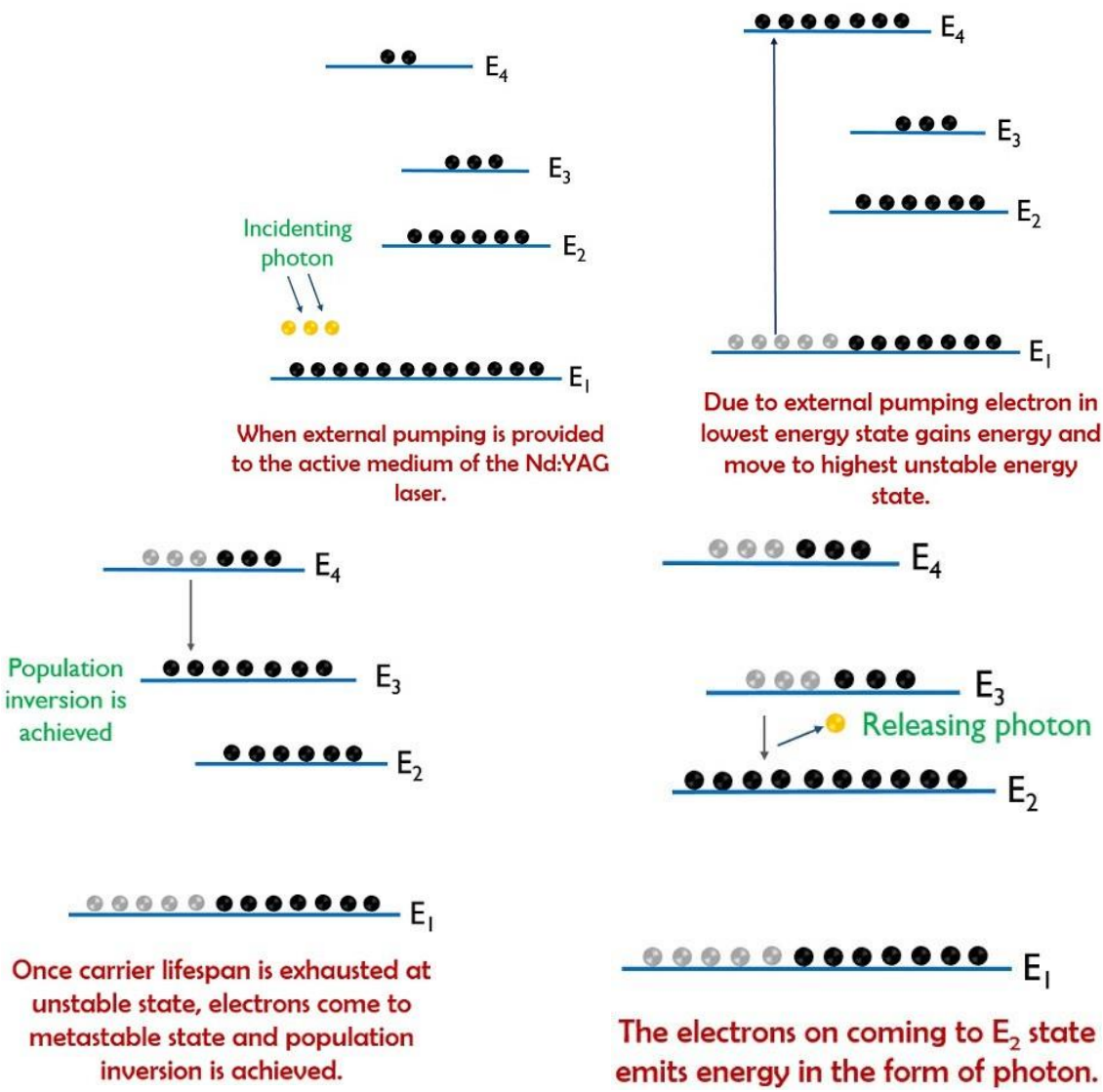


Fig. 2.1

Construction of Nd:YAG laser

- Active medium: when the external energy source is provided then the electrons from lower energy state moves to higher energy state thereby causing lasing action
- External Energy source: optical pumping, xenon or krypton flash tube is taken
- Nd:YAG rod and the flash tube are placed inside an elliptical cavity
- Optical resonator: two ends of the Nd:YAG rod is coated with silver. - to achieve maximum light reflection.
- other end is partially coated in order to provide a path for the light ray from an external source to reach the active medium.
- E_1 is the lowest energy state while E_4 is the highest energy level, electrons present in the energy state E_1 gains energy and moves to energy state E_4.E_4 is an unstable state.
- electrons that were excited to this state by the application external pumping will not stay at this state for much longer duration and comes to lower energy state E_3 very fastly but without radiating any photon.
- E_3 is the metastable state and exhibits longer lifespan. Thereby attaining population inversion.

- lifetime of the electrons at the metastable state gets exhausted then these electrons by releasing photons come to lower energy state E_2.
- E_2 also exhibit shorter lifespan like E_4. Thus, electrons present in E_2 state will come to E_1



Electrons by gaining single photon of energy releases the energy of 2 photons. Also, as the system is equipped with optical resonators so, more number of photons will get generated as the pumped energy will get reflected inside the active medium.

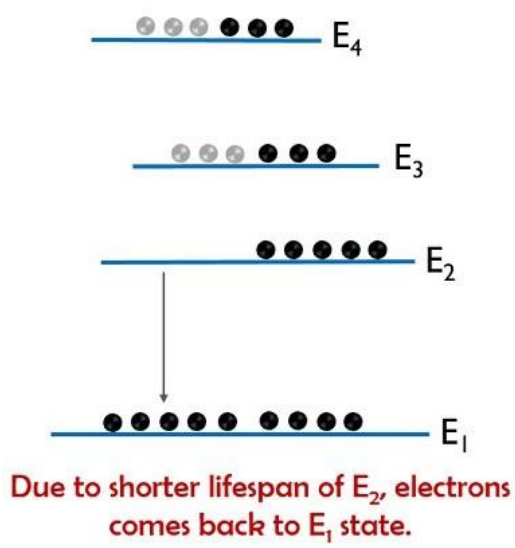


Fig. 2.2

several electrons on stimulation produce photons thereby generating a coherent laser beam of 1.064 µm.

Applications of Nd:YAG Laser

- Military applications to find the desired target.
- Application in medical field for the surgical purpose.
- Used in welding and cutting of steel and
- Used in communication system

Solid State Laser - Ruby Lser

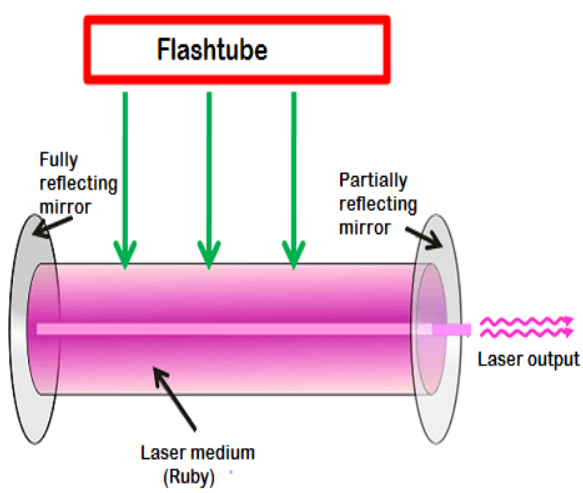


Fig. 2.3

Construction of Ruby laser

- Ruby is a crystal of aluminium oxide (Al_2O_3) in which some of the aluminium ions (Al^{3+}) are replaced by chromium ions (Cr^{3+}). This is done by doping small amounts of chromium oxide
- pink or red color depending upon the concentration of chromium ions
- Al_2O_3 does not participate in the laser action. It only acts as the host.
- Length of ruby crystal is usually 2 cm to 30 cm and diameter 0.5 cm to 2 cm.

High temperature is produced during the operation of the laser, the rod is surrounded by liquid nitrogen to cool

Active medium or active center: Chromium ions act as active centers in ruby crystal. So it is the chromium ions that produce the laser

Pumping source: A helical flash lamp filled with xenon is used as a pumping source. The ruby crystal is placed inside a xenon flash lamp. Thus, optical pumping is used to achieve population inversion in ruby laser.

Optical resonator system: The ends of ruby crystal are polished, grounded and made flat. The one of the ends is completely silvered while the other one is partially silvered to get the output. Thus the two polished ends act as optical resonator system.

Working

- ✓ Ruby is a three level laser system.
- ✓ there are three levels E_1, E_2 and (E_3 & E_4). E_1 is the ground level, E_2 is the metastable level, E_3 and E_4 are the bands. E_3 & E_4 are considered as only one level because they are very closed to each other.
- ✓ Pumping: The ruby crystal is placed inside a xenon flash lamp
- ✓ A part of this energy is absorbed by chromium ions in the ground state.
- ✓ optical pumping raises the chromium ions to energy levels inside the bands E_3 and E_4. This process is called stimulated absorption.

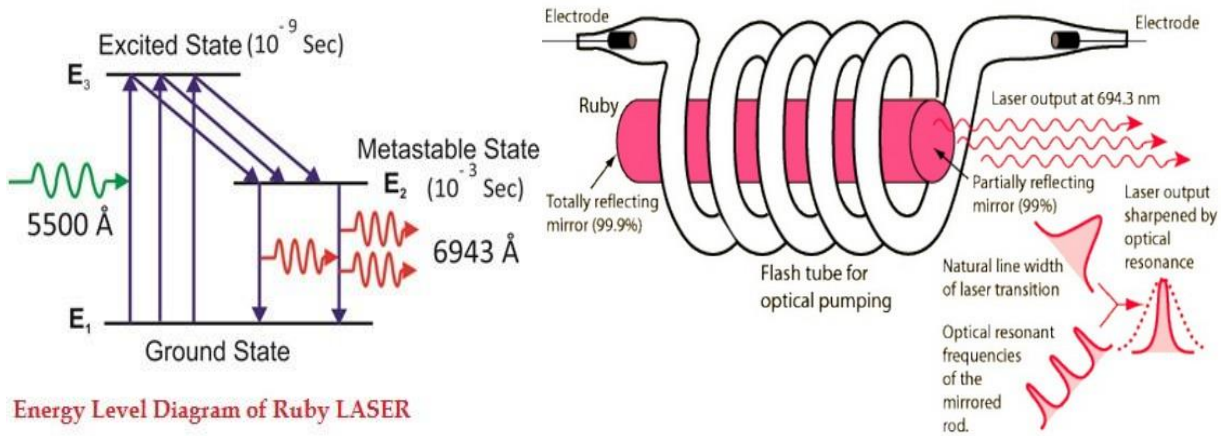


Fig. 2.4

Achievement of population inversion:

- Cr^{3+} ions in the excited state loose a part of their energy
- The transition from excited states to metastable state is non-radiative transition or in other words there is no emission of photons.
- The number of chromium ions goes on increasing in E_2 state, while due to pumping
- As a result, the number of chromium ions become more in excited state(metastable state) as compared to ground state E_1.
- Hence, the population inversion is achieved between states E_2 and E_1.
- Photon travels through the ruby rod and if it is moving in a direction parallel to the axis of the crystal, then it is reflected to and fro by the silvered ends of the ruby rod until it stimulates the other excited ions and cause it to emit a fresh photon in phase with the stimulating photon.
- Emitted photons will knock out more photons by stimulating the chromium ions and their total number sufficiently increases.

Output Measurement

In the energy level diagram, E2 is the upper laser level and E1 is the lower laser level because laser beam is achieved in between these levels. Thus, the ruby laser fits into the definition of three level laser system.

Output: The output wavelength of ruby laser is 6943 Å and output power is 10 raise to power 4 to 10 raise to power 6 watts and it is in the form of pulses.

Table 2.1

Type	Solid state laser; Three- level laser system
Active medium	Ruby rod (Cr: Al_2O_3)
Active centre	Cr^{3+} ion
Pumping method	Optical pumping
Pumping source	Helical flash lamp of filled with Xenon
Optical resonator	The ends of the ruby rod are kept in between two optically coated mirrors, silvered differently.
Output power	Low
Nature of the output	Pulsed (Spiked)
Wavelength emitted	693.4 nm

APPLICATIONS

1. Ruby laser has very high output power of the order of $10^4 – 10^6$ watts. It has wavelength of 6943 Angstroms.
2. Ruby lasers are used in industrial cutting and welding.
3. They are used for hair removal and tattoo
4. Holography, NDT, Decoration, Display and toys

GAS LASER

A gas laser is a type of laser in which a mixture of gas is used as the active medium or laser medium. Gas lasers are the most widely used lasers.

Gas lasers range from the low power helium-neon lasers to the very high power carbon dioxide lasers. commonly used in college laboratories whereas the carbon dioxide lasers are used in industrial applications.

The main advantage of gas lasers (eg: He-Ne lasers) over solid state lasers is that they are less prone to damage by overheating so they can be run continuously.

Helium-neon laser

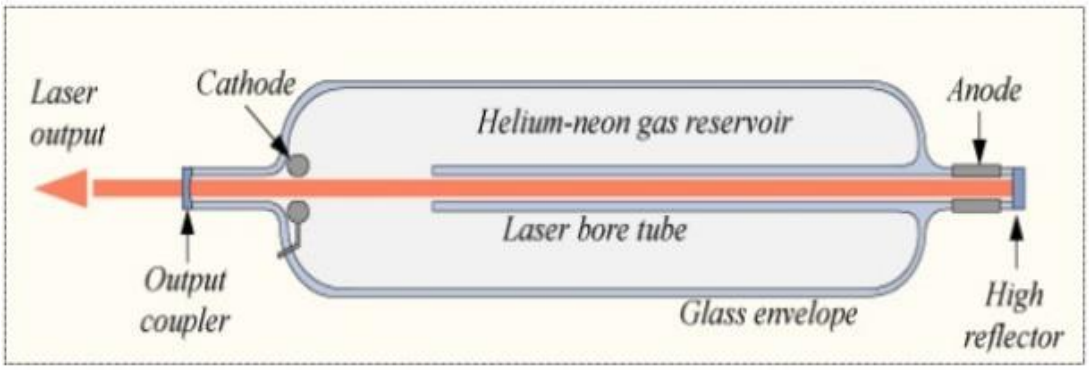


Fig. 2.5

- The helium-neon laser was the first continuous wave (CW) laser ever constructed
- The excitation of electrons in the He-Ne gas active medium is achieved by passing an electric current through the gas.
- The helium-neon laser operates at a wavelength of 632.8 nanometers (nm), in the red portion of the visible spectrum.

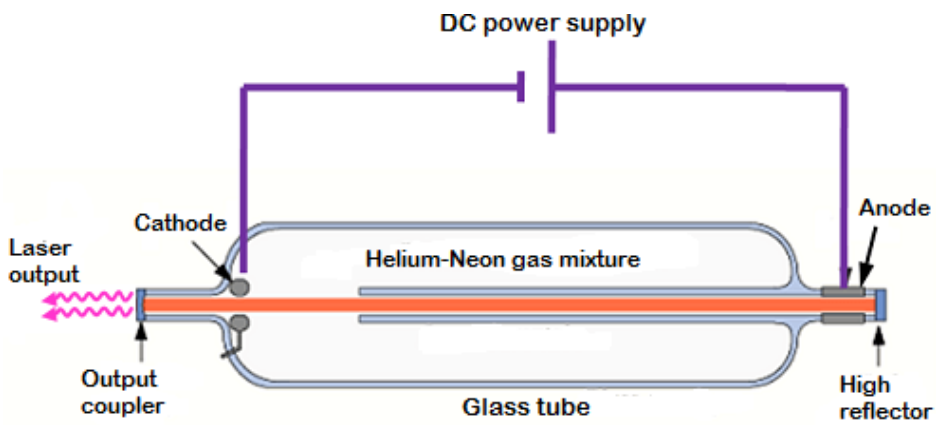


Fig. 2.6

Helium-neon laser construction

The helium-neon laser consists of three essential components:

- Pump source (high voltage power supply)
- Gain medium (laser glass tube or discharge glass tube)
- Resonating cavity
- High voltage power supply
- to achieve population inversion, we need to supply energy to the gain medium or active medium

- In helium-neon lasers, a high voltage DC power supply is used as the pump source. A high voltage DC supplies electric current through the gas mixture of helium and neon.

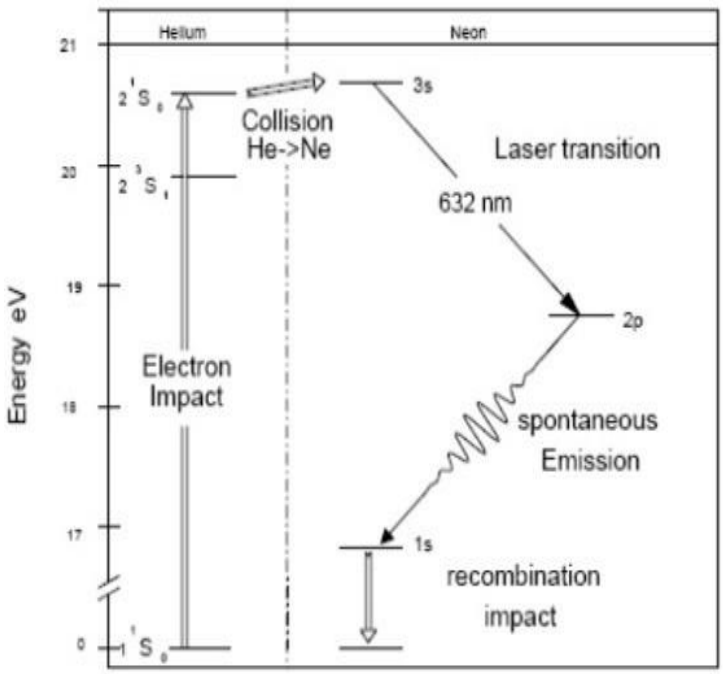


Fig. 2.7

Gain medium

- ✓ The partial pressure of helium is 1 mbar whereas that of neon is 0.1 mbar.
- ✓ to excite primarily the lower energy state electrons of the helium atoms.
- ✓ neon atoms are the active centers and have energy levels suitable for laser transitions while helium atoms help in exciting neon atoms.
- ✓ Electrodes (anode and cathode) are provided in the glass tube to send the electric current through the gas mixture. These electrodes are connected to a DC

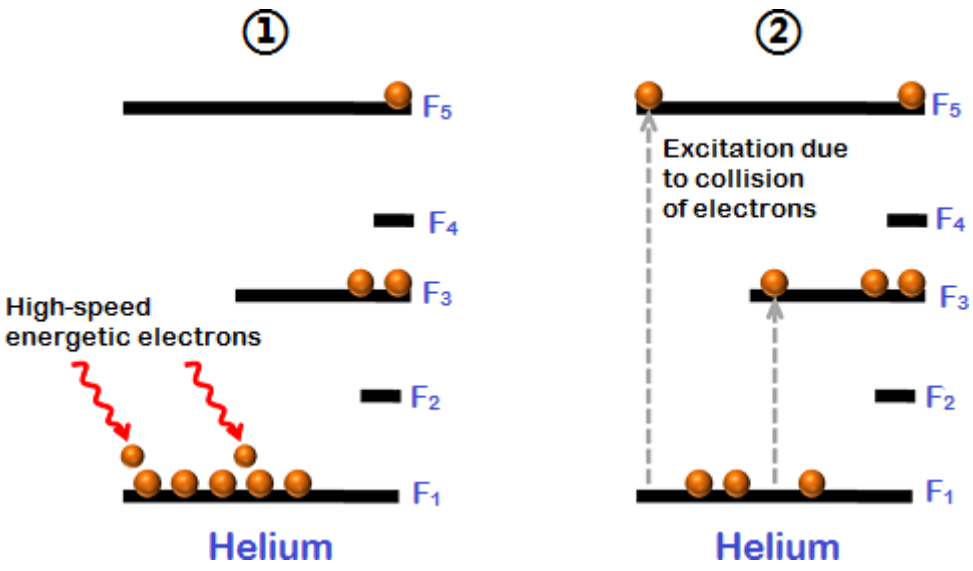
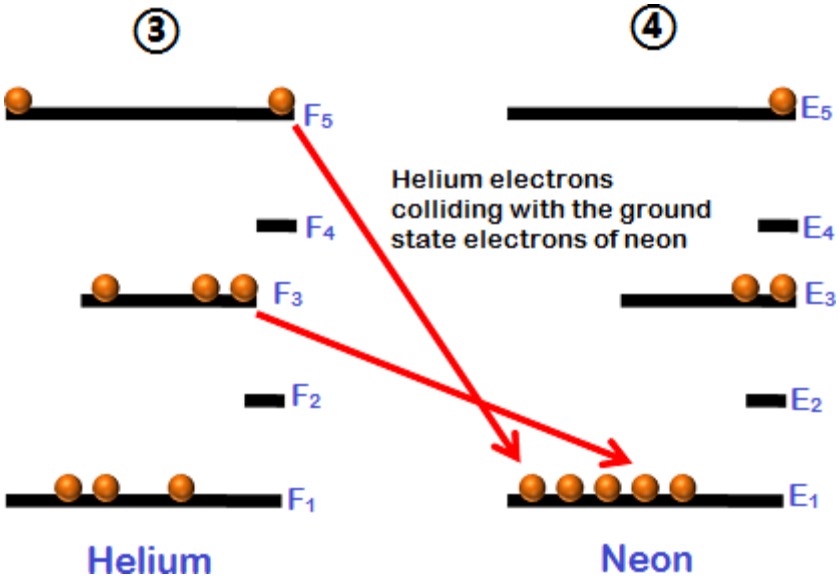


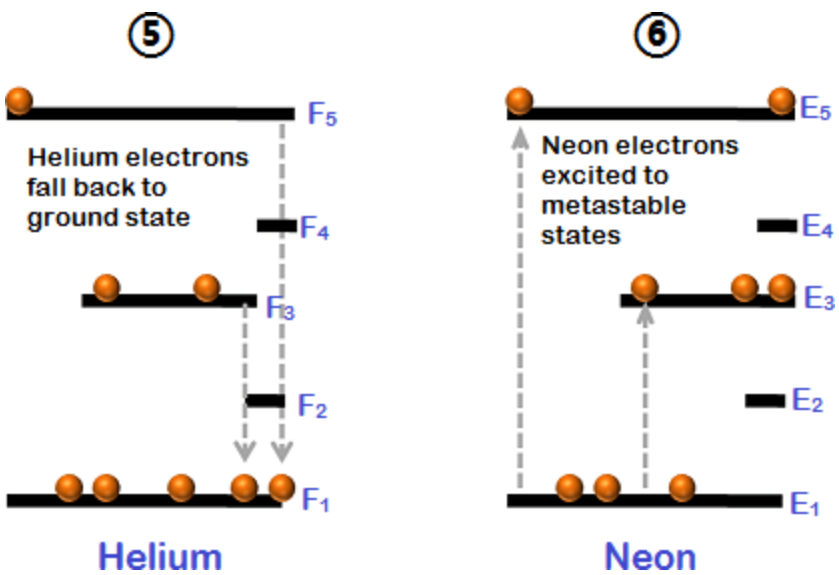
Fig. 2.1

the power is switched on, a high voltage of about 10 kV is applied It is enough to excite the electrons and are accelerated

- Electrons transfer some of their energy to the helium atoms, jumps into the excited states
- Assume that these metastable states are F3 and F5
- Metastable state electrons of the helium atoms, return to ground state by transferring their energy to the lower energy state electrons of the neon atoms.
- The energy levels of some of the excited states of the neon atoms are identical to the energy levels of metastable states of the helium atoms.
- Let us assume that these identical energy states are $F_3 = E_3$ and $F_5 = E_5$. E_3 and E_5 are excited states or metastable states of neon atoms.



the lower energy state electrons of the neon atoms gain enough energy from the helium atoms and jumps into the higher energy states or metastable states (E_3 and E_5) whereas the excited electrons of the helium atoms will fall into the ground state. Thus, helium atoms help neon atoms in achieving population inversion.



- millions of ground state electrons of neon atoms are excited to the metastable states having longer lifetime
- electrons (E_3 and E_5) of the neon atoms will spontaneously fall into the next lower energy states (E_2 and E_4) by releasing photons or red light.
- Neon excited electrons continue on to the ground state through radiative and nonradiative transitions.

- Photons emitted from the neon atoms will moves back and forth between two mirrors until it stimulates other electrons
- optical gain is achieved due to stimulated emission

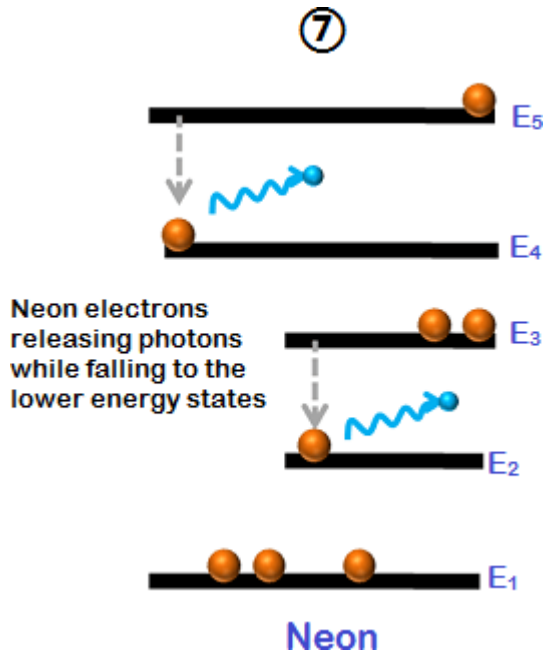


Fig. 2.8

photons emitted will escape through the partially reflecting mirror or output coupler to produce laser.

Advantages of helium-neon laser

- Helium-neon laser emits laser light in the visible portion of the spectrum.
- High stability
- Low cost
- Operates without damage at higher temperatures

Disadvantages of helium-neon laser

- Low efficiency
- Low gain
- Helium-neon lasers are limited to low power tasks

Applications of helium-neon lasers

- Helium-neon lasers are used in industries.

- Helium-neon lasers are used in scientific instruments.
- Helium-neon lasers are used in the college laboratories

MOLECULAR GAS LASER -CO_2

CO_2 Molecular gas laser: Principle, Construction, Working, Characteristics, Advantages, Disadvantages and Applications

In a molecular gas laser, laser action is achieved by transitions between vibrational and rotational levels of molecules. Its construction is simple and the output of this laser is continuous.

Molecular Gas laser

In a molecular gas laser, laser action is achieved by transitions between vibrational and rotational levels of molecules. Its construction is simple and the output of this laser is continuous.

In CO2 molecular gas laser, transition takes place between the vibrational states of Carbon dioxide molecules.

CO2 **Molecular gas laser**

It was the first molecular gas laser developed by Indian born American scientist Prof.C.K.N.Pillai.

It is a four level laser and it operates at 10.6 µm in the far IR region. It is a very efficient laser.

Energy states of CO2 molecules.

A carbon dioxide molecule has a carbon atom at the center with two oxygen atoms attached, one at both sides. Such a molecule exhibits three independent modes of vibrations. They are

a) Symmetric stretching mode.

b) Bending mode

c) Asymmetric stretching mode.

a. **Symmetric stretching mode**

In this mode of vibration, carbon atoms are at rest and both oxygen atoms vibrate simultaneously along the axis of the molecule departing or approaching the fixed carbon atoms.

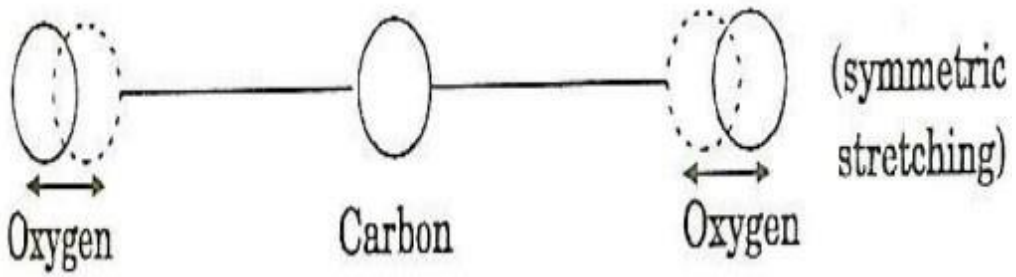


Fig. 2.9

b. **Bending mode:**

In this mode of vibration, oxygen atoms and carbon atoms vibrate perpendicular to molecular axis.

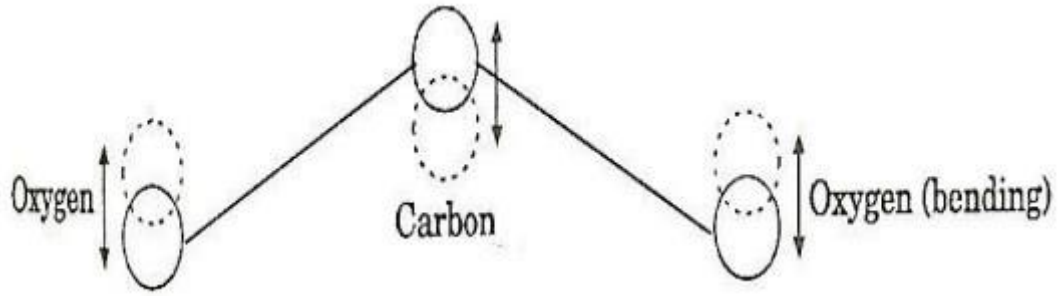


Fig.2.10 2.1

c. **Asymmetric stretching mode**:

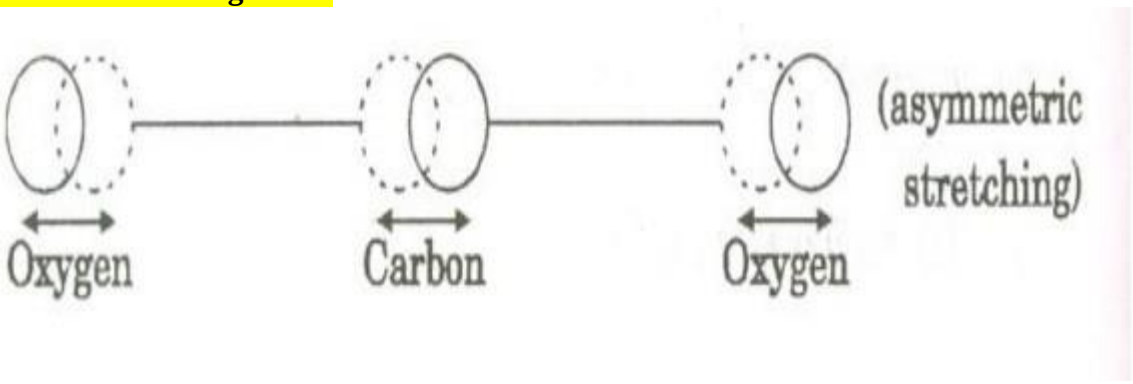


Fig. 2.11

In this mode of vibration, oxygen atoms and carbon atoms vibrate asymmetrically, i.e., oxygen atoms move in one direction while carbon atoms in the other direction.

Principle:

The active medium is a gas mixture of CO2, N2 and He. The laser transition takes place between the vibrational states of CO2molecules.

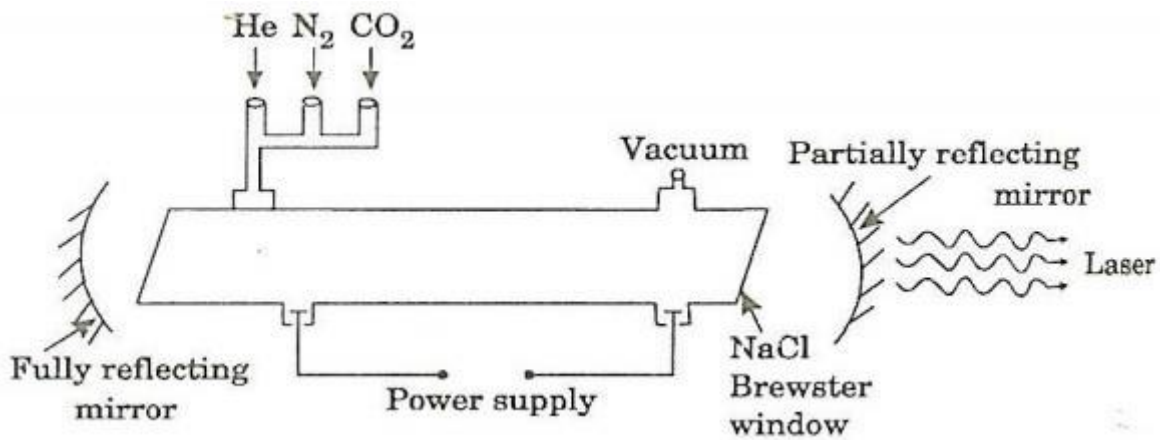


Fig. 2.12

Construction:

It consists of a quartz tube 5 m long and 2.5 cm in the diameter. This discharge tube is filled with gaseous mixture of CO2(active medium), helium and nitrogen with suitable partial pressures.

The terminals of the discharge tubes are connected to a D.C power supply. The ends of the discharge tube are fitted with NaCl Brewster windows so that the laser light generated will be polarized.

Two concave mirrors one fully reflecting and the other partially form an optical resonator.

Working:

Figure shows energy levels of nitrogen and carbon dioxide molecules.

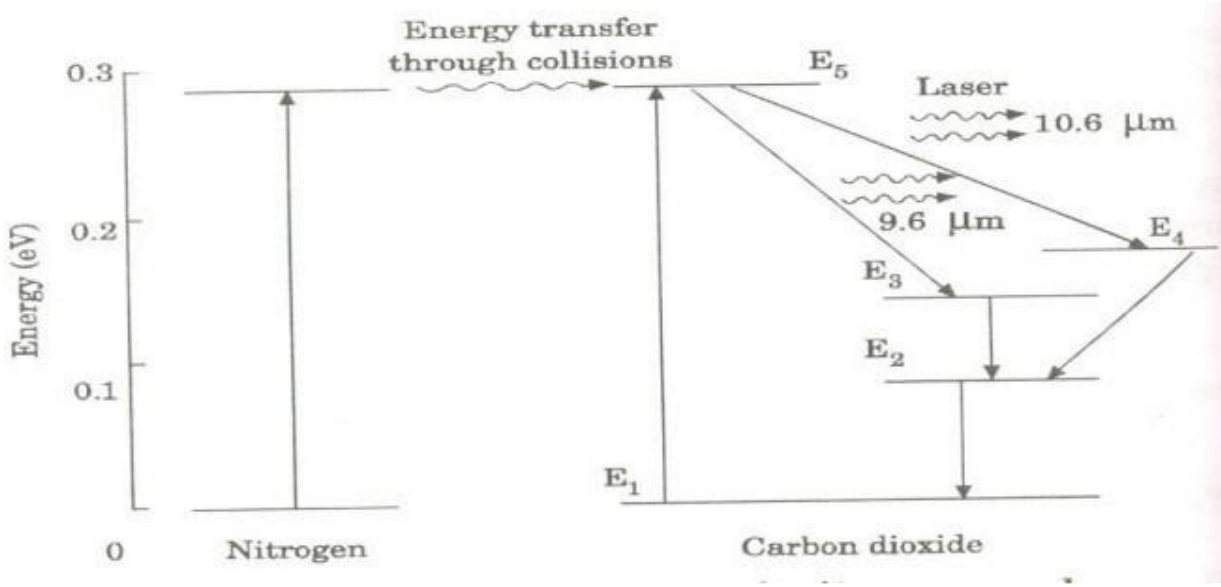


Fig. 2.13

When an electric discharge occurs in the gas, the electrons collide with nitrogen molecules and they are raised to excited states. This process is represented by the equation

$N_2 + e^* = N_2^* + e$

N_2 = Nitrogen molecule in ground state e* = electron with kinetic energy

N_2^* = nitrogen molecule in excited state e= same electron with lesser energy

Now N_2 molecules in the excited state collide with CO2 atoms in ground state and excite to higher electronic, vibrational and rotational levels.

This process is represented by the equation N2* + CO2 = CO2* + N2

N2* = Nitrogen molecule in excited state. CO2 = Carbon dioxide atoms in ground state

CO2* = Carbon dioxide atoms in excited state N2 = Nitrogen molecule in ground state.

Since the excited level of nitrogen is very close to the E5 level of CO2 atom, population in E5 level increases.

As soon as population inversion is reached, any of the spontaneously emitted photon will trigger laser action in the tube. There are two types of laser transition possible.

1. **Transition E5 to E4** :

This will produce a laser beam of wavelength 10.6μm

2. **Transition E5 to E3** This transition will produce a laser beam of wavelength 9.6μm. Normally 10.6μm transition is more intense than 9.6μm transition. The power output from this laser is 10kW.

Characteristics:

1. Type: It is a molecular gas laser.

2. Active medium: A mixture of CO_2 , N_2 and helium or water vapour is used as active medium
3. Pumping method: Electrical discharge method is used for Pumping action

4. Optical resonator: Two concave mirrors form a resonant cavity

5. Power output: The power output from this laser is about 10kW.

6. Nature of output: The nature of output may be continuous wave or pulsed wave.

7. Wavelength of output: The wavelength of output is 0.6µm and 10.6µm.

Advantages:

1. The construction of CO_2 laser is simple

2. The output of this laser is continuous.

3. It has high efficiency

4. It has very high output power.

5. The output power can be increased by extending the length of the gas tube.

Disadvantages:

1. The contamination of oxygen by carbon monoxide will have some effect on laser action

2. The operating temperature plays an important role in determining the output power of laser.

3. The corrosion may occur at the reflecting plates.

4. Accidental exposure may damage our eyes, since it is invisible (infra red region) to our eyes.

Applications:

1. High power CO_2 laser finds applications in material processing, welding, drilling, cutting soldering etc.

2. The low atmospheric attenuation (10.6µm makes CO_2 laser suitable for open air communication.
3. It is used for remote sensing

4. It is used for treatment of liver and lung diseases.

5. It is mostly used in neuro surgery and general surgery.

6. It is used to perform microsurgery and bloodless operations.

SEMICONDUCTOR LASER

Laser action can also be produced semiconductors. The most compact of all the lasers in semiconductor diode laser. It is also called injection laser. There are two types of semiconductor diode lasers (***i.) Homo - junction laser (ii.) Hetero- Junction laser.***

HOMO – JUNCTION SEMICONDUCTOR DIODE LASER

Definition

It is specifically fabricated p-n junction diode. This diode emits laser light when it is forward biased.

Principle

When a p-n junction diode is forward biased, the electrons from n – region and the holes from the p- region cross the junction and recombine with each other. During the recombination process, the light radiation (photons) is released from a certain specified direct band gap semiconductors like Ga-As. This light radiation is known as recombination radiation.

The photon emitted during recombination stimulates other electrons and holes to recombine. As a result, stimulated emission takes place which produces laser.

Construction

Figure shows the basic construction of semiconductor laser. The active medium is a p-n junction diode made from the single crystal of gallium arsenide. This crystal is cut in the form of a platter having thickness of 0.5µm. The platelet consists of two parts having an electron conductivity (n-type) and hole conductivity (p-type).

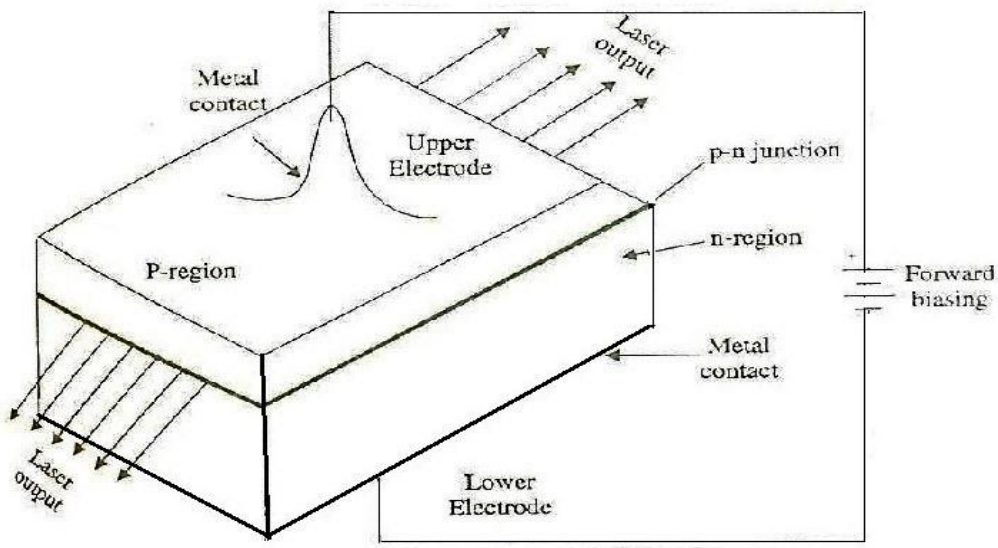


Fig. 2.14

The photon emission is stimulated in a very thin layer of PN junction (in order of few microns). The electrical voltage is applied to the crystal through the electrode fixed on the upper surface. The end faces of the junction diode are well polished and parallel to each other. They act as an optical resonator through which the emitted light comes out.

Working

Figure shows the energy level diagram of semiconductor laser. When the PN junction is forward biased with large applied voltage, the electrons and holes are injected into junction region in considerable concentration. The region around the junction contains a large amount of electrons in the conduction band and a large amount of holes in the valence band.

If the population density is high, a condition of population inversion is achieved. The electrons and holes recombine with each other and this recombination's produce radiation in the form of light. When the forward – biased voltage is increased, more and more light photons are emitted and the light production instantly becomes stronger.

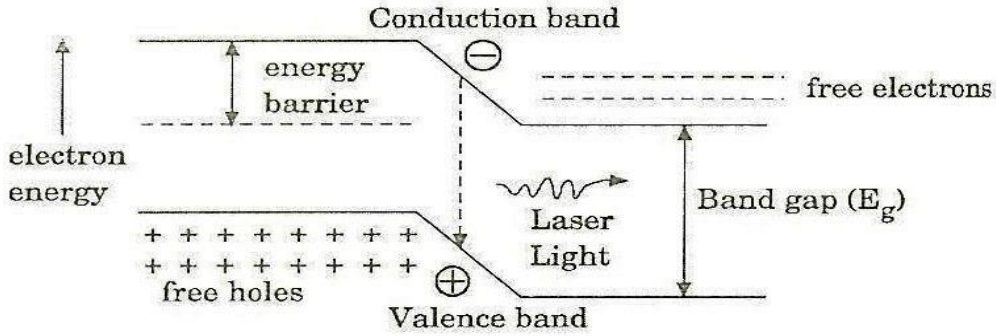


Fig. 2.15

These photons will trigger a chain of stimulated recombination resulting in the release of photons in phase.

The photons moving at the plane of the junction travels back and forth by reflection between two sides placed parallel and opposite to each other and grow in strength.

After gaining enough strength, it gives out the laser beam of wavelength $8400A^0$

The wavelength of laser light is given by $E_g = h\nu = h\frac{c}{\lambda}$

$\lambda = \frac{hc}{E_g}$ where Eg is the band gap energy in joule.

Characteristics

1. ***Type:*** It is a solid state semiconductor laser.
2. ***Active medium:*** A PN junction diode made from single crystal of gallium arsenide is used as an active medium.
3. ***Pumping method:*** The direct conversion method is used for pumping action

4. ***Power output:*** The power output from this laser is a few mW.
5. ***Nature of output:*** The nature of output is continuous wave or pulsed output.
6. ***Wavelength of Output***: gallium arsenide laser gives infrared radiation in the wavelength 8300 to $8500A^0$.

Advantages:

1. It is very small in dimension. The arrangement is simple and compact.
2. It exhibits high efficiency.
3. The laser output can be easily increased by controlling the junction current
4. It is operated with lesser power than ruby and CO_2 laser.
5. It requires very little auxiliary equipment
6. It can have a continuous wave output or pulsed output.

Disadvantages

1. It is difficult to control the mode pattern and mode structure of laser.
2. The output is usually from 5° to 15° i.e., laser beam has large divergence.
3. The purity and monochromacity are poor than other types of laser
4. Threshold current density is very large ($400A/mm^2$).
5. It has poor coherence and poor stability.

Application

1. It is widely used in fiber optic communication
2. It is used to heal the wounds by infrared radiation
3. It is also used as a pain killer
4. It is used in laser printers and CD writing and reading

HETERO - JUNCTION SEMICONDUCTOR DIODE LASER

A p-n junction made up of the different materials in two regions ie., n type and p type is known as Hetero junction.

Principle:

When a PN junction diode is forward biased, the electrons from the n region and holes from the p region recombine with each other at the junction. During recombination process, light is released from certain specified direct band gap semiconductors.

Construction:

This laser consists of five layers as shown in the figure. A layer of Ga-As p – type (3rd layer) will act as the active region. This layer is sand witched between two layers having wider band gap viz. Ga Al As-p – type (2nd layer) and Ga Al As-n- type (4th layer). The end faces of the junctions of 3rd and 4th layer are well polished and parallel to each other. They act as an optical resonator.

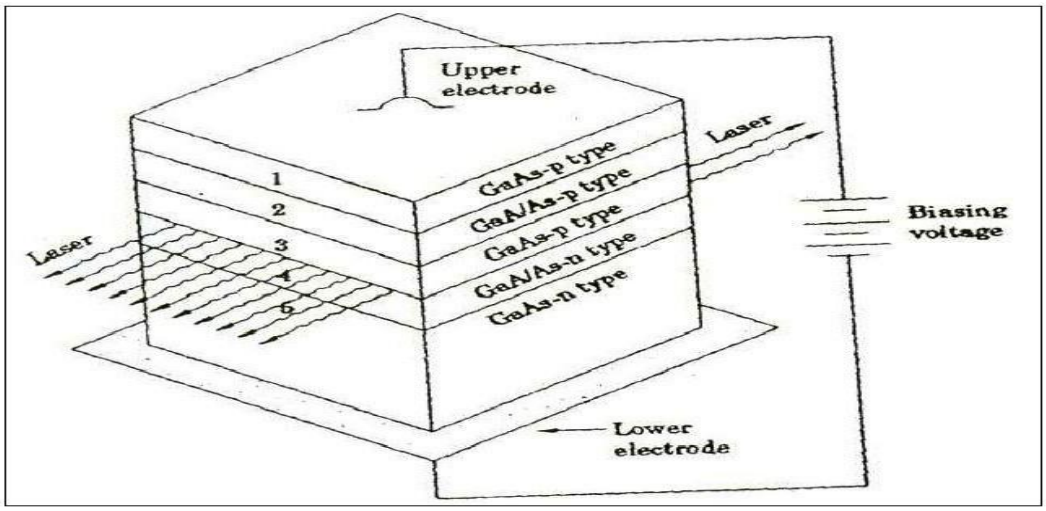


Fig. 2.16

Working:

When the PN junction is forward biased, the electrons and holes are injected into the junction region. The region around the junction contains large amount of electrons in the conduction band and holes in the valence band. Thus the population inversion is achieved. At

this stage, some of the injected charge carriers recombines and produce radiation in the form of light.

When the forward biased voltage is increased, more and more light photons are emitted and the light intensity is more. These photons can trigger a chain of stimulated recombination's resulting in the release of photons in phase.

The photons moving at the plane of the junction travels back and forth by reflection between two sides and grow its strength. A coherent beam of laser having wavelength nearly $8000A^0$ emerge out from the junction region.

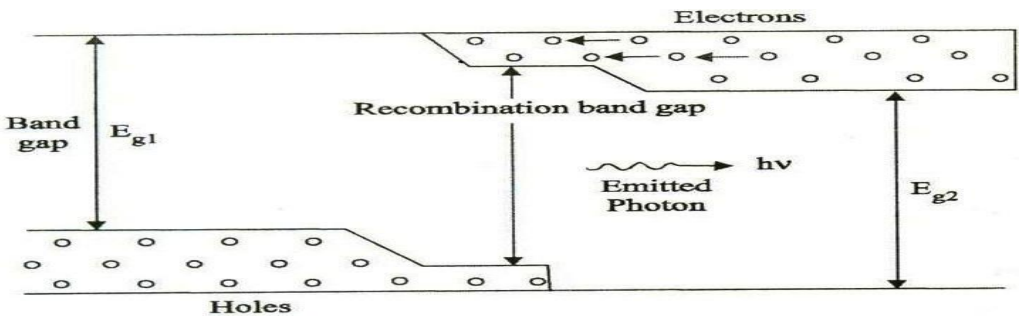


Fig. 2.17

Characteristics:

1. ***Type:*** It is a Hetero junction semiconductor laser
2. ***Active medium:*** PN junctions made from different layers.
3. ***Pumping method:*** Direct conversion method
4. ***Power output:*** The power output of laser beam is 1 mW
5. ***Nature of the Output:*** Continuous wave form
6. ***Wavelength of the output:*** Nearly 8000 A^0

Advantages:

1. It produces continuous wave output.
2. The power output is very high.

Disadvantages:

1. It is very difficult to grow different layers of PN junction.
2. The cost is very high.

Applications:

1. This type of laser is mostly used in optical applications
2. It is widely used in computers, especially on CD-ROMs.

LIQUID DYE LASER

Dye is a liquid laser with an active medium
Doped with Rhodamine B, Rhodamine 6G, fluoresein

Characteristics

- Output lies in UV, Visible or IR
- Using dye the output can be varied from 390nm to 1000nm
- Power output starts from 1 watt, beam diameter of 0.5mm
- Conversion efficiency is relatively high than 25%

Construction

It has two configuration

- The dye is pumped through the capillary tube from the storage tank
- It gets optically excited by the flash tube
- Output pass through the Brewster window

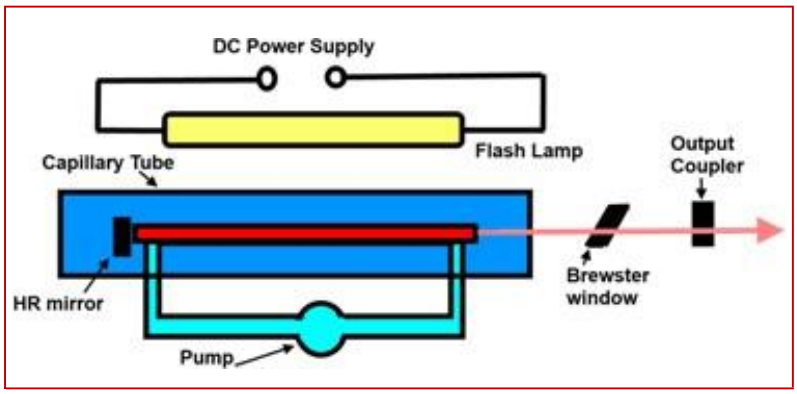


Fig. 2.18

2nd Configuration

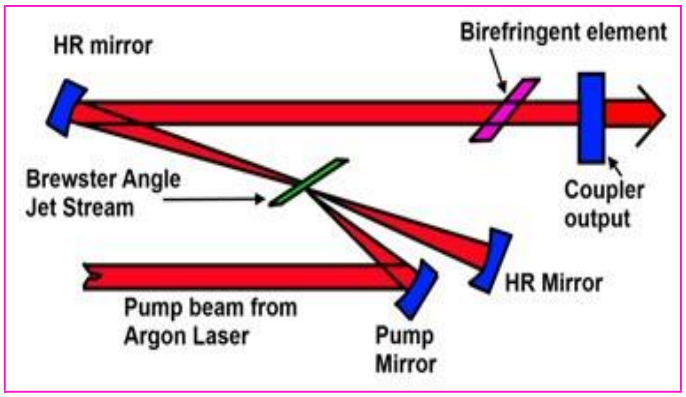


Fig. 2.19
The dye is pumped through the nozzle to form a Brewster Angle
Excitation mechanism involves here
Laser gets reflected from two HR mirror passed through the output coupler
Birefringent filter is used to tune the output of Laser

Working

- Active medium in Laser may be of organic dye mixed with ethanol, water
- Dyes like Rhodamine B, Sodium fluorisin it's difficult to determine the element
- So organic dye is prefered
- It provides the output in various ranges of wavelength
- The amount of amplification also varies
- Birefringent filter acts as a prism
- Filters certain specified frequency, it bends the other wavelength, tune the output with accuracy

Applications
It is used as research tool in medical Applications

Advantages

- It is available in visible and non-visible form
- Construction of a dye laser is not so complex
- Beam dia. is less
- Ranges of wavelength can be gained as output
- Having greater efficiency
- High power output is possible

Disadvamtages

- Cost of dye laser is high
- Using filter as birefringent makes it costly
- To determine the particular element of output, a complex dye has to be used

Q SWITCHING

Q switching is a technique for obtaining energetic short (but ot ultrashot) pulses from a laser by modulating the intracavity losses and thus the Q-factor of the laser resonator. The technique is mainly applied for the generation of nanosecond pulses of high energy and peak power with solid-state bulk lasers.

Generation of a Q-switched pulse

- The resonator losses are kept at a high level - As lasing cannot occur, gain medium enhanced by the pumping mechanism - spontaneous emission
- The losses are suddenly (with active or passive means) reduced
- The power of the laser radiation builds up very quickly in the laser resonator
- The saturation energy of the gain medium, starts to be saturated.
- When the gain equals the loss, peak of the pulse is reached
- The pulse duration achieved with Q switching is typically in the nanosecond range
- The energy of the generated pulse is typically higher than the energy of the gain medium
- In most cases, Q-switched lasers generate regular pulse trains via repetitive Q switching.
- Q-switched lasers have reached pulse durations far below 1 ns and repetition rates up to several megahertz
- laser systems can deliver pulses with many kilojoules of energy and durations in the nanosecond range.
- The first experimental demonstrations were performed in 1961 at Hughes Aircraft Company
- The resonator losses can basically be switched in different ways:

Active Q Switching
Passive Q Switching

In order to store many atoms in an upper level, the flow to a lower level must first be limited.
Thus, stimulated emission must be prevented by placing an attenuator in the cavity to stop light from travelling back and forth (note: this attenuator is usually a light modulator, rather than a mechanical shutter, which reduces the amplitude or power of the light beam).
In this case, for a radiative transition, the only decay to a lower level is due to spontaneous emission. When the pumping system supplies more atoms per second than lose energy by spontaneous emission, the population in the upper level can become very large

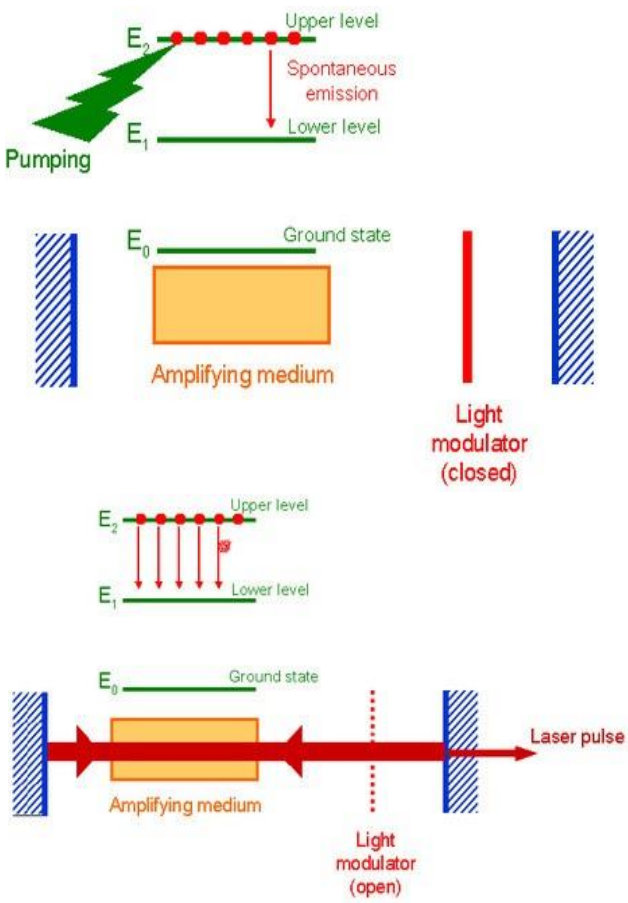


Fig. 2.20

- ✓ After a certain time, the energy losses in the cavity are suddenly reduced so that laser oscillation becomes possible.
- ✓ As there is a very large population in the upper level, stimulated emission becomes very probable and the laser is suddenly triggered.
- ✓ The flow due to stimulated emission is much greater than the other flows (filling by pumping and emptying by spontaneous emission): all the atoms stored in the upper level fall sharply, emitting stimulated photons (starting with the spontaneous emission trapped in the cavity).
- ✓ Thus, the laser cavity fills with stimulated photons at the same time as the upper level empties
- ✓ Eventually, the upper level is completely empty.
- ✓ There is no further stimulated emission and the cavity will also empty due to the losses created by the output mirror (in general, the cavity empties after only a few round trips)

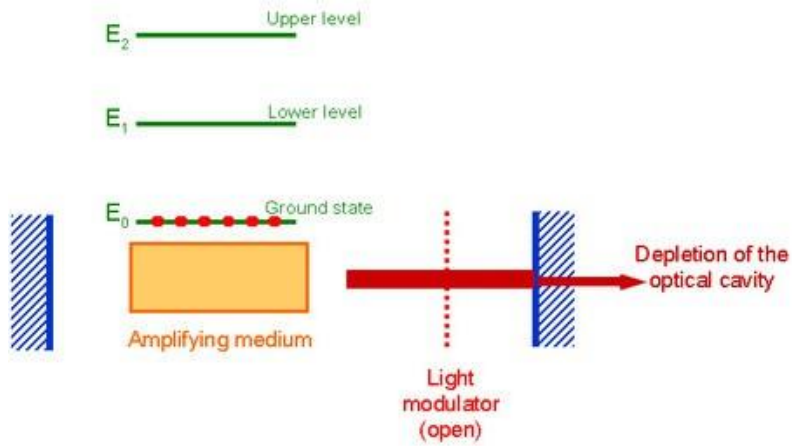


Fig. 2.21

- ✓ This process gives rise to a dramatic variation in the number of photons in the cavity (first by a significant amplification due to stimulated emission then by the complete emptying of the cavity at the end).
- ✓ The net result is the emission of a short pulse of light via the output mirror.
- ✓ Generally, several round trips are needed to completely depopulate the upper energy level and several more round trips to empty the optical cavity so the duration of the pulse is greater than one round trip.
- ✓ This means that for optical cavities shorter than a metre, it is possible to generate short pulses of only a few nanoseconds but several millijoules in power.
- ✓ The peak power (the pulse energy divided by its duration) of these lasers can be in the megawatt range or even higher.
- ✓ It should be noted that Q-switched lasers never reach a steady state as they stop functioning after several round trips of the light in the cavity.

Active Q Switching

For active Q switching, the losses are modulated with an active control element (active Q switch), typically either an acousto-optic or electro-optic modulator.

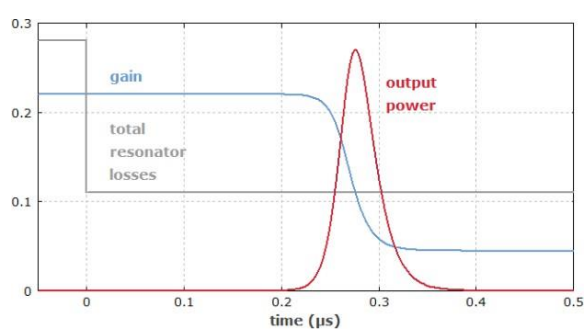


Fig. 2.22

the switching time of the modulator is not comparable to the pulse duration
For many applications, Q-switched pulses are generated in a periodic fashion, i.e., with a given pulse repetition rate.

Passive Q Switching

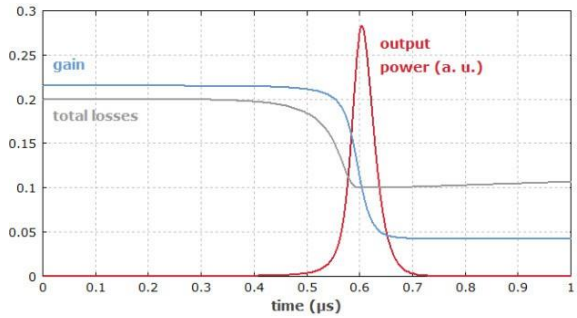


Fig. 2.23

MODE-LOCKING

- Operating technique is completely different.
- The cavity is filled with photons everywhere at the same time: only a packet of photons is allowed to propagate in the cavity.
- This pulse lasts for a shorter time than a round trip in the cavity. In other words, its spatial extension is markedly shorter than the length of the cavity.
- operating conditions consists a light modulator, that can chop the light in the cavity into periods of exactly the same length as a round trip.
- Those photons allowed to pass through the modulator in its on-state will be amplified

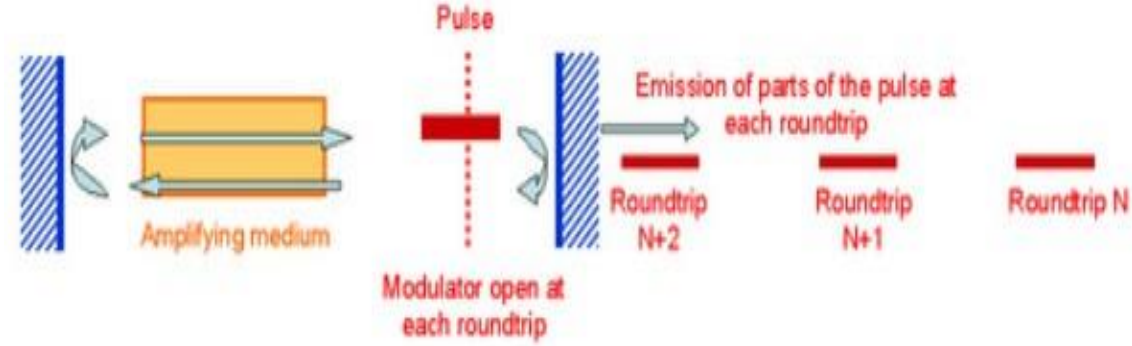


Fig. 2.24 *A pulse propagating in the optical cavity of a mode-locked laser*

- It reaches the modulator in this state after each round trip.
- The other photons elsewhere in the cavity will be subject to losses when they travel through the modulator.
- The pulses last for a much shorter time than a round trip in the cavity.
- photons allowed to pass, will be amplified
- the pulses last for a much shorter time than a round trip in the cavity.

- They are limited by the Fourier transform of the spectrum emitted by the laser: the wider the spectrum, the shorter the pulse.
- exceptionally wide - pulse generated will be only several femtoseconds long
- The pulse repetition period corresponds to the cavity round-trip time
- The average power of a mode-locked laser is of the same order of magnitude as that of continuous-wave lasers.
- In fact, in contrast to Q-switched lasers, these can also reach a steady state like continuous-wave lasers.
- The fundamental difference is that the stimulated photons are condensed in a packet rather than spread all around the cavity.
- During one round trip, only one laser pulse is emitted via the output mirror.
- The pulse energy is thus equal to the average power multiplied by the duration of a round trip.
- Generally, these energies are of the order of several nanojoules.
- The term “mode-locking” comes from the analysis of the various frequencies.
- A laser operating under these conditions will emit over several different frequencies due to the rapid modulation of the modulator.
- if the laser emits continuously at two frequencies separated by , the light output due to interference of the two waves will be modulated by a sinusoidal term of frequency
- modulation is generally very rapid, detected by photodiodes
- known as beating and results from the interference of beams with different frequencies
- When a large number of frequencies are emitted by the laser, the beat signal becomes quite complex.
- Its shape depends on the relative phase of the waves with different frequencies.
- However, the beat signal has a very regular shape in one particular case: when all the waves emitted by the cavity are in phase.
- Then, there are certain times and spots in the cavity where all the waves beat in phase and the interference signal is thus very powerful

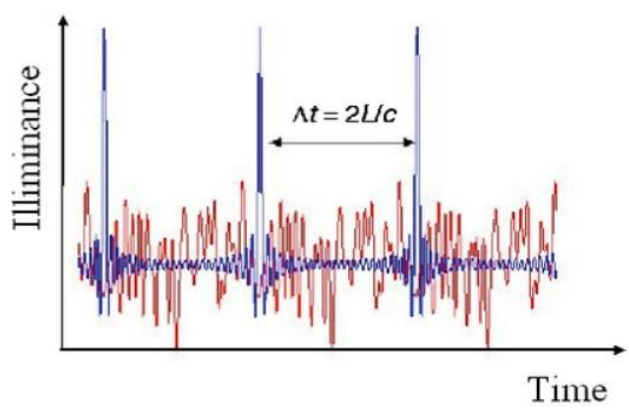


Fig. 2.25

- When the longitudinal modes are in phase, in the cavity where the electric fields add together constructively.
- Everything occurs as if a pulse was travelling inside the cavity

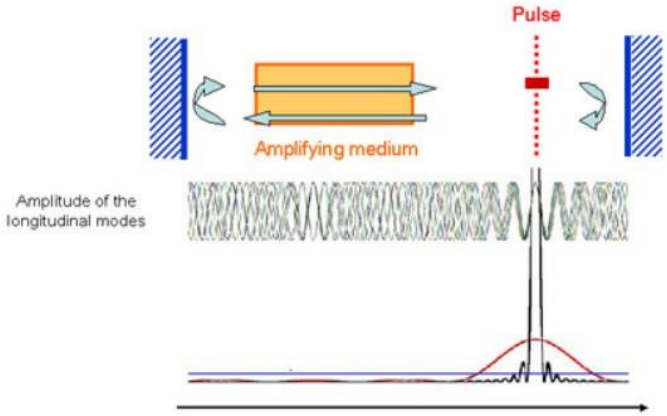


Fig. 2.1

LASERS MODES: THE SHAH FUNCTION

The Shah function, III(t), is an infinitely long train of equally spaced delta-functions. The symbol III is pronounced *shah* after the Cyrillic character III, which is said to have been modeled on the Hebrew letter (shin) which, in turn, may derive from the Egyptian a hieroglyph depicting papyrus plants along the Nile.

The Fourier transform of the Shah function

$$= \int_{-\infty}^{\infty} \sum_{m=-\infty}^{\infty} \delta(t-m) \exp(-i\omega t) dt$$

$$= \sum_{m=-\infty}^{\infty} \int_{-\infty}^{\infty} \delta(t-m) \exp(-i\omega t) dt$$

$$= \sum_{m=-\infty}^{\infty} \exp(-i\omega m)$$

If $w = 2np$, where n is an integer, every term is
exp(-2mnp i) = 1, and the sum diverges; otherwise, cancellation occurs. So:

$$\boxed{\Upsilon\ \{\mathrm{III}(t)\} \propto \mathrm{III}(\omega/2\pi)}$$

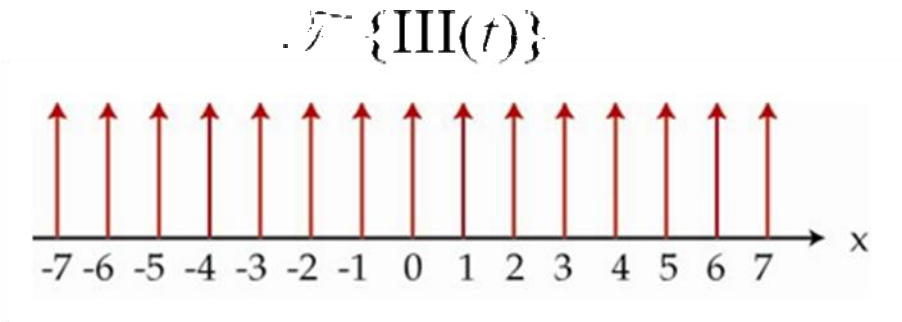


Fig. 2.1 The Fourier transform of an infinite train of pulses

An infinite train of identical pulses can be written:

$E(t) = III(t/T) * f(t)$

where *f(t)* represents a single pulse and *T* is the time between pulses.

The Convolution Theorem states that the Fourier Transform of a convolution is the product of the Fourier Transforms. So:

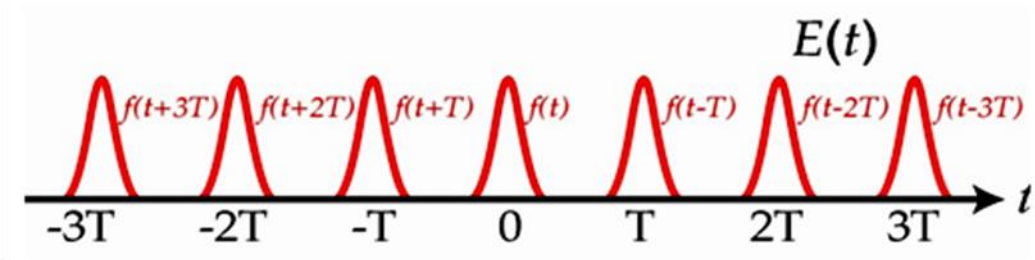


Fig. 2.1

$$\tilde{E}(\omega) \propto III(\omega T / 2\pi) F(\omega)$$

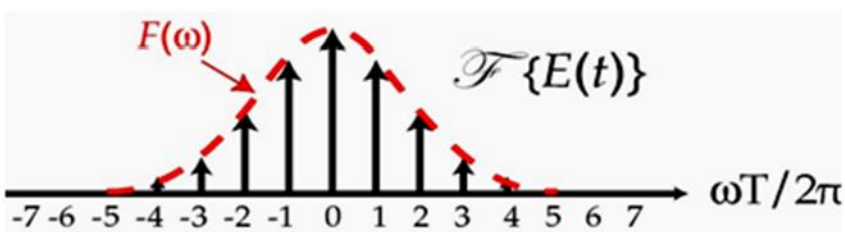


Fig. 2.1

A train of pulses results from a single pulse bouncing back and forth inside a laser cavity of round-trip time *T*. The spacing between frequencies—called laser modes—is then dw = 2p/*T* or δ**n = 1/*T***.

MODE-LOCKED VS. NON-MODE-LOCKED LIGHT

Mode-locked pulse train:

$$\tilde{E}(\omega) = \sum_{m=-\infty}^{\infty} F(\omega)\,\delta(\omega - 2\pi m/T)$$

Random phase for each mode

$$= F(\omega) \sum_{m=-\infty}^{\infty} \delta(\omega - 2\pi m/T) = F(\omega)\,\mathrm{III}(\omega T/2\pi)$$

$$\tilde{E}(\omega) = \sum_{m=-\infty}^{\infty} F(\omega)\exp(i\varphi_m)\,\delta(\omega - 2\pi m/T)$$

$$= F(\omega) \sum_{m=-\infty}^{\infty} \exp(i\varphi_m)\,\delta(\omega - 2\pi m/T)$$

Generating short pulses = mode-locking

Locking the phases of the laser modes yields an ultrashort pulse.

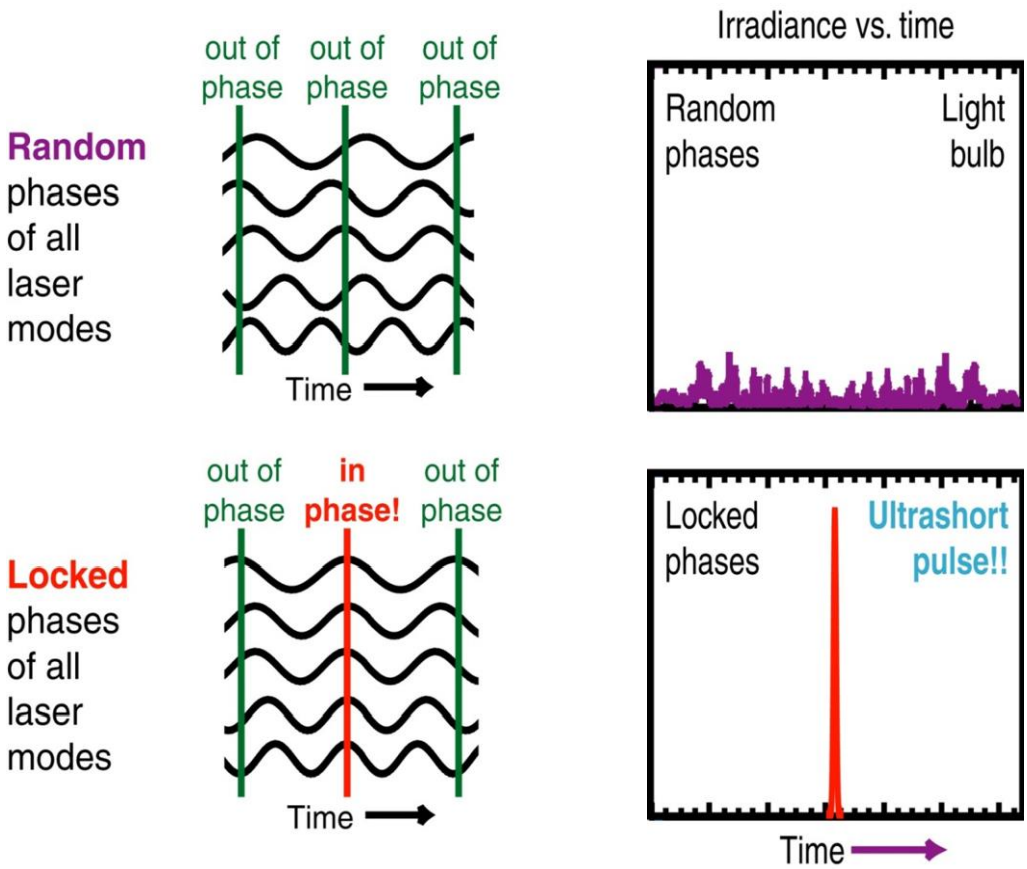


Fig. 2.1

Locked modes

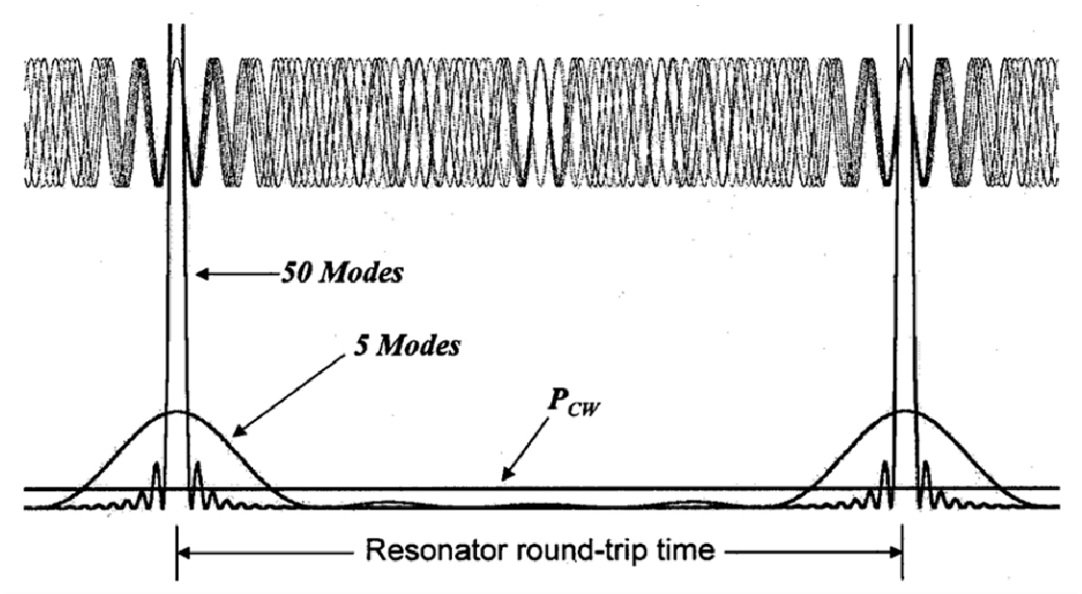


Fig. 2.1

Numerical simulation of mode-locking

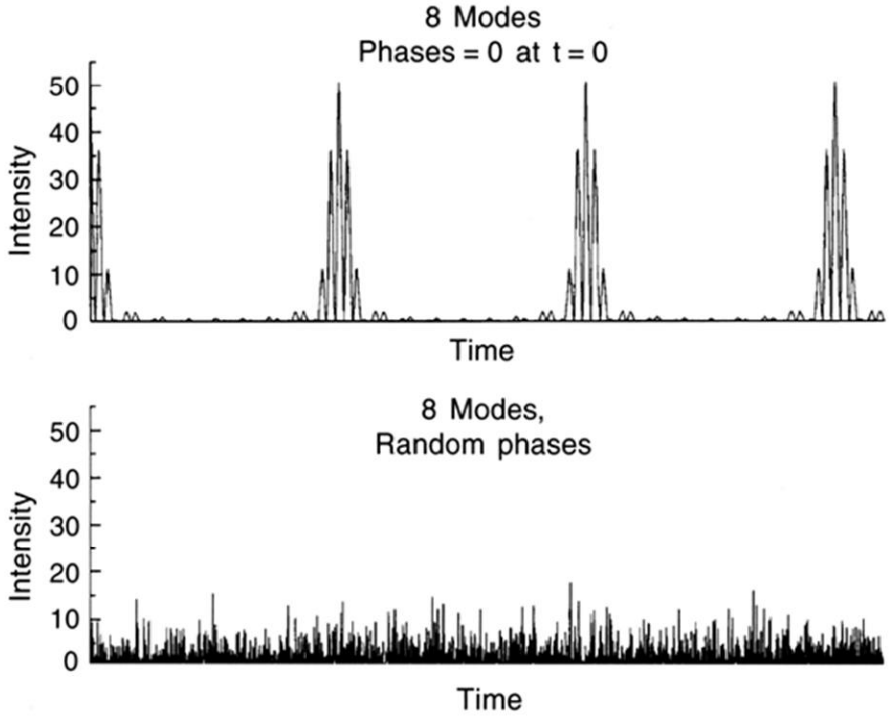


Fig. 2.1 Ultrafast lasers often have thousands of modes.

A generic ultrashort-pulse laser

A generic ultrafast laser has a broadband gain medium, a pulse-shortening device, and two or more mirrors:

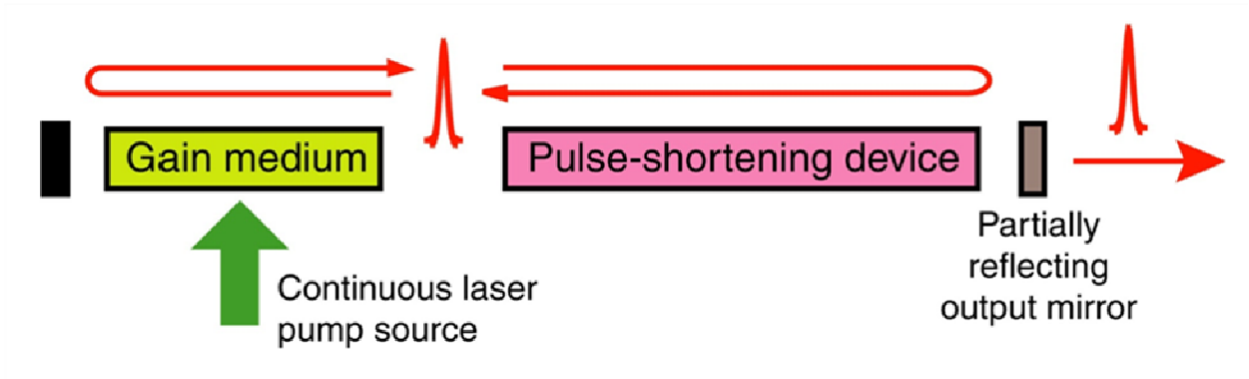


Fig. 2.1 Many pulse-shortening devices have been proposed and used.

Passive mode-locking: the saturable absorber

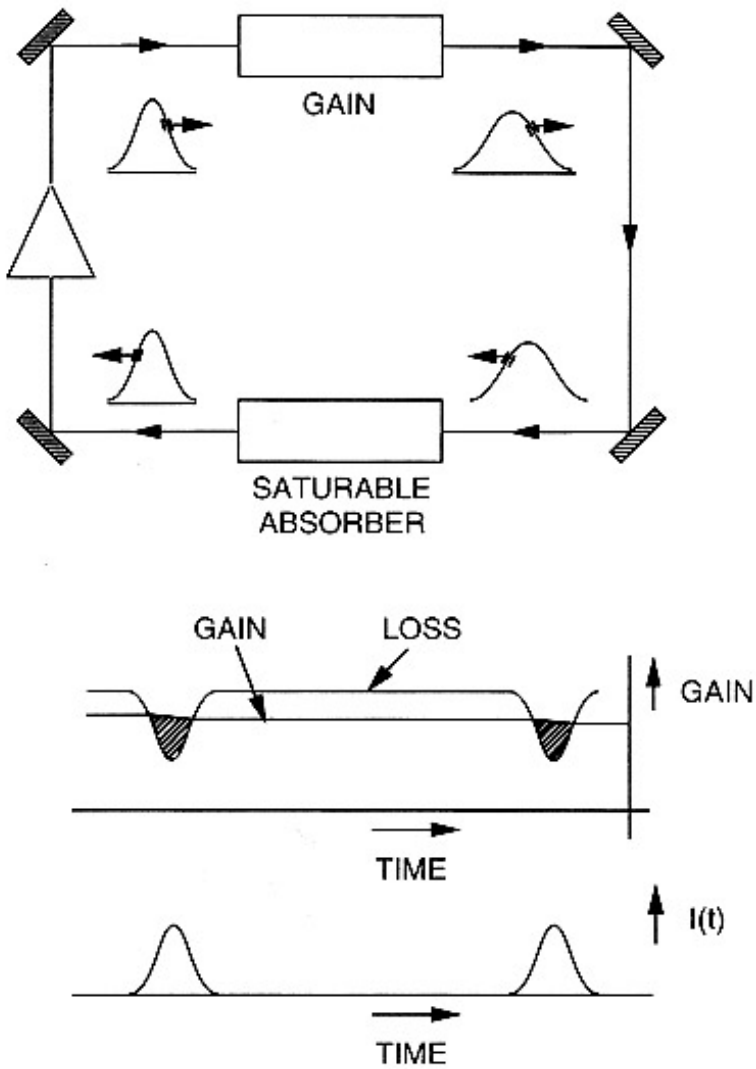


Fig. 2.1

High-intensity spikes (i.e., short pulses) see less loss and hence can lase while low-intensity backgrounds (i.e., long pulses) won't.

Passive mode-locking with a slow saturable absorber

- What if the absorber responds slowly (more slowly than the pulse)?
- Then only the leading edge will experience pulse shortening.

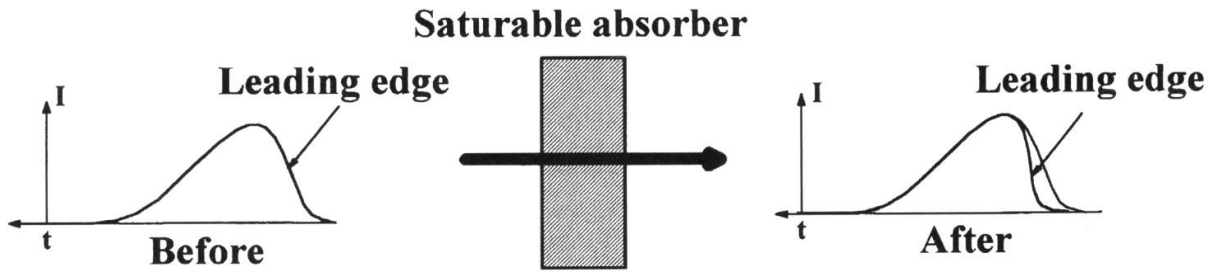


Fig. 2.1

This is the most common situat

The Passively Mode-locked Dye Laser

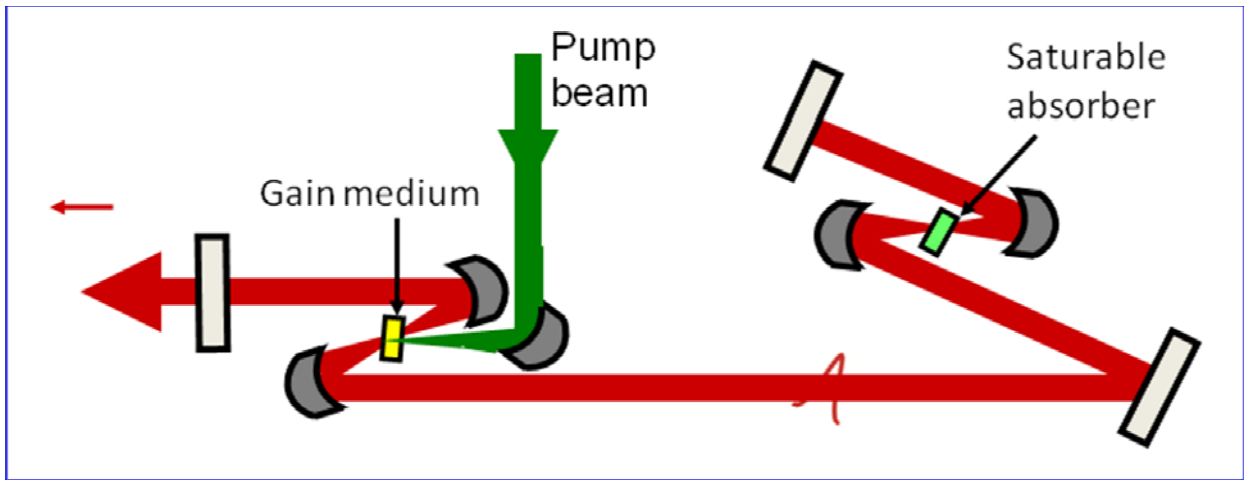


Fig. 2.1

- Passively mode-locked dye lasers yield pulses as short as a few hundred fs.
- They're limited by our ability to saturate the absorber.

Commercial fs lasers - Availability

- Ti:Sapphire
- Coherent:
- Mira (<35 fs pulse length, 1 W ave power),
- Chameleon (Hands-free, ~100 fs pulse length),
- Spectra-Physics:
- Tsunami (<35 fs pulse length, 1 W ave power)

- Mai Tai

(Hands-free, ~100 fs

pulse length) Active

mode-locking

✓ Any amplitude modulator can preferentially induce losses for times other than that of the
intended pulse peak. This produces short pulses.

✓ It can be used to start a

Ti:Sapphire laser mode-locking

Hybrid mode-locking

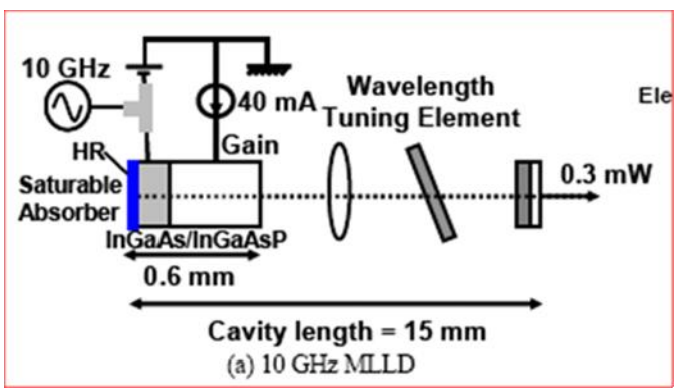
Hybrid mode-locking is any type of mode-locking incorporating two or more techniques simultaneously.

✓ Sync-pumping and passive mode-locking

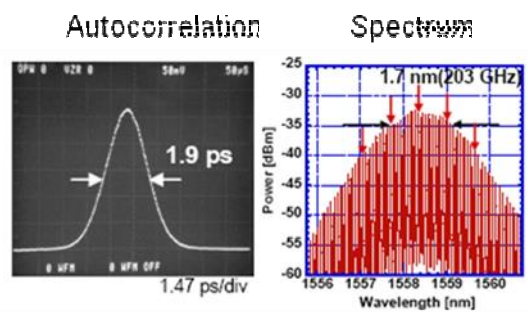
✓ Active and passive mode-locking

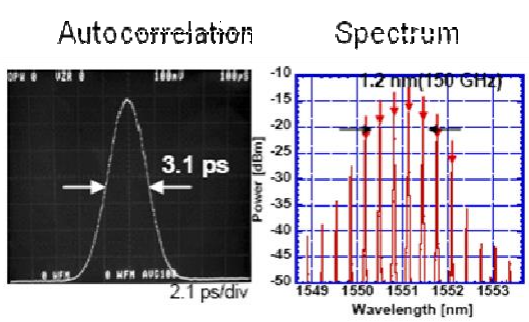
However, using two lousy methods together doesn't really work all that much better than one good method.

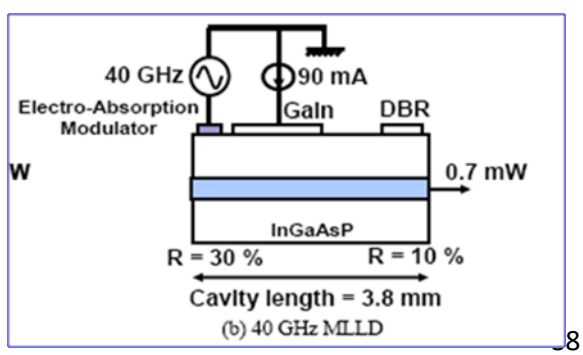
Diode lasers use hybrid mode-locking



(a) 10 GHz MLLD







(b) 40 GHz MLLD

Fig. 2.1